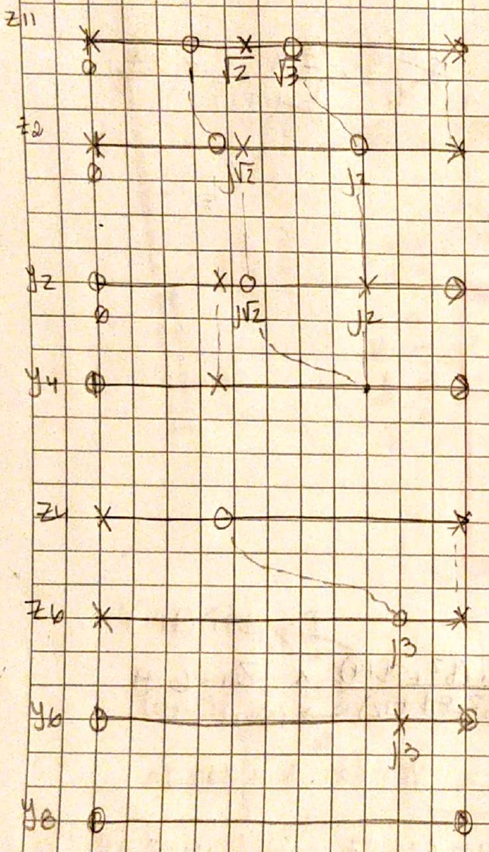
9) sintetizar con un circuito escalera

$$T(s) = \frac{(s^2+4)(s^2+9)}{(s^2+1)(s^2+3)} \Rightarrow T = \frac{V_2}{V_1} = \frac{Z_{21}}{Z_{11}} = \frac{-Y_{21}}{Y_{22}}$$

$$= \frac{\dfrac{(s^2+4)(s^2+9)}{s(s^2+2)}}{\underbrace{\dfrac{(s^2+1)(s^2+3)}{s(s^2+2)}}_{Z_{11}}}$$

Tenemos que usar las funciones de excitación que sí conocemos como sintetizar
↓
Necesitamos los ceros de T

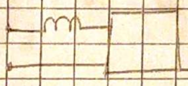


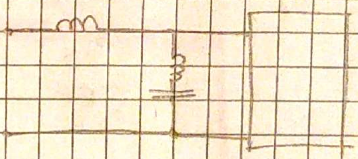
removemos de forma parcial

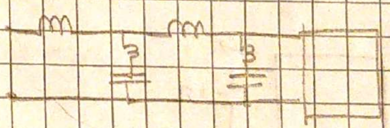
al remover un polo finito, el cero que llega deja un valor reactivo

nuevamente de forma parcial

removemos el polo en admitancia







$$\rightarrow Z_2 = Z_{11} - K_\infty s \quad \wedge \quad Z_2\Big|_{s^2=-4} = Z_{11}\Big|_{s^2=-4} - K'_\infty \cdot s = 0 \quad \text{con} \quad K'_\infty = \frac{Z_{11}}{s}\Big|_{s^2=-4} = \frac{(s^2+1)(s^2+3)}{s(s^2+2)\,s}\Big|_{-4}$$

$$Z_2 = \frac{(s^2+1)(s^2+3)}{s(s^2+2)} - \frac{3}{8}s = \frac{s^4+4s^2+3-\frac{3}{8}s^2(s^2+2)}{s(s^2+2)} = \frac{\frac{5}{8}s^4+\frac{13}{4}s^2+3}{s(s^2+2)} \qquad K'_\infty = \frac{3}{8} \qquad Z_2 = \frac{\frac{5}{8}(s^2+\frac{6}{5})(s^2+4)}{s^3+2s}$$

$$\rightarrow Y_2 = \frac{s(s^2+2)}{\frac{5}{8}(s^2+\frac{6}{5})(s^2+4)} \Rightarrow Y_4 = Y_2 - \frac{K \cdot s}{s^2+4} \quad \text{con} \quad K = \lim_{s^2\to-4} \frac{s^2+4}{s} \cdot Y_2 = \lim_{s^2\to-4} \frac{s^2+4}{s} \cdot \frac{s(s^2+2)}{\frac{5}{8}(s^2+\frac{6}{5})(s^2+4)}$$

$$K = \lim_{s^2\to-4} \frac{s^2+2}{\frac{5}{8}(s^2+\frac{6}{5})} = \frac{-2}{-\frac{7}{4}} = \frac{8}{7}$$

$$Y_4 = \frac{s(s^2+2)}{\frac{5}{8}(s^2+\frac{6}{5})(s^2+4)} - \frac{\frac{8}{7}\cdot s}{(s^2+4)}$$

$$= \frac{s(s^2+2) - \frac{8}{7}s\,\frac{5}{8}(s^2+\frac{6}{5})}{\frac{5}{8}(s^2+\frac{6}{5})(s^2+4)} = \frac{s^3+2s-\frac{5}{7}s^3-\frac{6}{7}s}{\frac{5}{8}(s^2+\frac{6}{5})(s^2+4)} = \frac{\frac{2}{7}s^3+\frac{8}{7}s}{\frac{5}{8}(s^2+\frac{6}{5})(s^2+4)}$$

$$\Rightarrow Z_4 = \frac{\frac{5}{8}(s^2+\frac{6}{5})(s^2+4)}{\frac{2}{7}s^3+\frac{8}{7}s} = \frac{\frac{5}{8}(s^2+\frac{6}{5})(s^2+4)}{\frac{2}{7}(s^3+4s)}$$

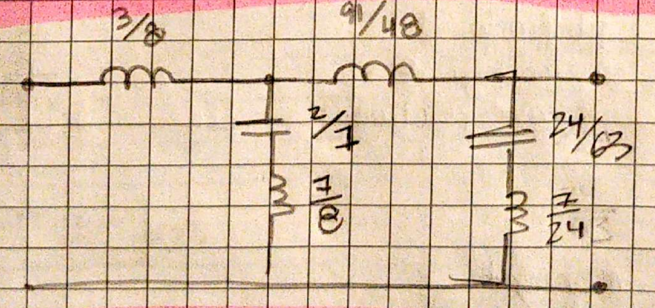
$$\Rightarrow Z_6 = Z_4 - k'_\infty s \quad \text{con} \quad Z_4\Big|_{s^2=-9} - k'_\infty s = 0 \Rightarrow Z_4\Big|_{s^2=-9} = k'_\infty \cdot s$$

$$\text{con } k'_\infty = \frac{Z_4}{s}\Big|_{s^2=-9} = \frac{\frac{5}{8}(s^2+\frac{6}{5})\cancel{(s^2+4)}}{\frac{2}{7}s\cancel{(s^2+4)}}\cdot\frac{1}{s}\Big|_{s^2=-9} = \frac{\frac{5}{8}(-9+\frac{6}{5})}{\frac{2}{7}(-9)} = \frac{-\frac{39}{8}}{-\frac{18}{7}} = \frac{91}{48}$$

$$Z_6 = \frac{\frac{5}{8}(s^2+\frac{6}{5})}{\frac{2}{7}s} - \frac{91}{48}\cdot s = \frac{\frac{5}{8}(s^2+\frac{6}{5}) - \frac{91}{48}s\cdot\frac{2}{7}s}{\frac{2}{7}s} = \frac{\frac{1}{12}s^2+\frac{3}{4}}{\frac{2}{7}s} = \frac{\frac{1}{12}(s^2+9)}{\frac{2}{7}s}$$

$$\Rightarrow y_6 = \frac{\frac{2}{7}s}{\frac{1}{12}(s^2+9)} \rightarrow y_8 = y_6 - \frac{ks}{s^2+9} \quad \text{con} \quad k = \lim_{s^2\to -9} \frac{\cancel{s^2+9}}{\cancel{s}}\cdot\frac{\frac{2}{7}\cancel{s}}{\frac{1}{12}\cancel{(s^2+9)}} = \frac{24}{7}$$

$$y_8 = \frac{\frac{2}{7}s}{\frac{1}{12}(s^2+9)} - \frac{\frac{24}{7}s}{(s^2+9)} = \frac{\frac{2}{7}s - \frac{24}{7}s\cdot\frac{1}{12}}{\frac{1}{12}(s^2+9)} = 0$$



8) Sintetizar cuadripolo RC con ... $(s+1)(s+4)$